GitHub and AI’s Influence: Technosocial Utopia or Dystopia?

AI Product Governance & Socio-Technical Systems Analysis

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1 System Context

GitHub is a widely used collaborative platform for building and shipping software. Since its release in 2008, GitHub has exploded in popularity, with over one hundred million users. Described by the company as "the world's most widely adopted AI-powered developer platform" (GitHub, n.d.), it helps users code, plan, automate, and secure their work. A diverse community of individuals and businesses uses GitHub for professional software development, education, social interaction, creative endeavors, and more. Many prominent companies utilize GitHub, including Duolingo, Mercedes-Benz, and Mercado Libre. GitHub is such a popular platform thanks to its fast speeds, high security, and AI-powered tools. Copilot, the AI assistant, helps companies streamline programming and develop software more efficiently. GitHub's rapid innovation through Copilot reflects the broader AI hype trend, with enthusiastic claims of a future of growth, efficiency, and productivity. However, these promises fail to capture the detrimental effects of environmental harm, the decline of human intelligence, and the reinforcement of existing social hierarchies.

2 Background / History

GitHub was formed in 2007 by Chris Wanstrath, P.J. Hyett, Tom Preston-Werner, and Scott Chacon, a group of American software developers. It was created as a platform for Git, a free and open-source tool released in 2005. Git offers high speed, minimal disk space usage, data integrity, flexibility, and ease of learning. As a central hub for Git development, GitHub is a platform for Git's distributed version control system (DVCS), meaning anyone can access the complete repository, facilitating collaboration. The platform's main features include code review, project management, associated web pages (GitHub Pages), and continuous integration unit testing. Continuous integration unit testing is like a document's version history, allowing you to ensure the code still works after changes.

When Microsoft bought GitHub in 2018, it sparked concern among developers. The major for-profit corporation has previously faced antitrust issues for monopolistic practices that could limit competitors. GitHub users worried about Microsoft's influence on the open-source community. However, this acquisition was part of a larger shift under CEO Satya Nadella, who took over in 2014. Nadella transitioned the company toward a greater commitment to open-source systems. After its GitHub purchase, the company shifted its focus beyond the Windows operating system to the Linux operating system and cloud services such as Copilot and Azure (Evans & DeBacker, 2020). Azure, Microsoft's cloud computing platform, features OpenAI services and integrates with GitHub and other open-source platforms to enhance its offerings (Microsoft, n.d.). Microsoft is making strides towards an "AI-powered future" by providing its partners with expanded AI skill training, benefit packages, product enhancements, and official certifications for AI expertise (Dezen, 2024). Through these developments, Microsoft has paved the way for the adoption of AI in its services and has transformed GitHub into a central platform for AI innovation and large-scale applications.

3 Connection to Artificial Intelligence and Machine Learning: AI Hype

GitHub offers AI assistance to users through its Copilot feature. GitHub's Copilot analyzes code to suggest edits and improve it. It "manages the tedious tasks" to streamline the coding process and is advertised as "AI that builds with you" and "Your code's guardian angel" (GitHub, 2025). It is portrayed as a helpful, collaborative assistant that still keeps the user in the driver's seat. Framing the tool in this way appeals to users because they would likely feel more competent if they still feel they are doing the work themselves. Users want to feel capable and accomplished, so Copilot's marketing targets their best interests, making them feel they are obtaining AI help while maintaining self-competence.

In *AI Snake Oil*, Narayanan and Kapoor argue that bold claims about AI's benefits, or "AI hype," often do not align with its actual capabilities. It is easier and more profitable for companies to advertise flawed AI products as powerful and perfect, rather than being transparent and explicit about shortcomings (Narayanan & Kapoor, 2024). Sociologist Ruha Benjamin explains how monopolists pursue a narrative of AI as our savior, catering to our needs and desires and optimizing human potential, breaking the boundaries of human capabilities. Benjamin argues that developers of AI are "futurists [letting their] imagination run wild," pushing the boundaries of technical capabilities while their "visions grow limp" when it comes to transforming society and improving lives (Benjamin, 2023). GitHub Copilot's glorifying, uncritical advertising reflects this techno-utopian narrative, referring to AI as a "guardian angel." It is portrayed as a valuable, beneficial tool, while its flaws are vastly omitted. This is part of the AI hype trend, in which AI is uncritically acclaimed, and its capabilities are exaggerated.

GitHub also contains many open-source AI and machine learning projects. For example, the platform is home to a top project, *AutoGPT*, an "experimental autonomous AI agent" (Asay, 2025). Generative AI projects are now among the most popular on the platform, representing a new emphasis on AI within open-source software development. GitHub and other open-source platforms support the rapid growth of AI and machine learning projects by providing useful frameworks, including cloud infrastructure and DevOps automation. DevOps is the practice of using technology to automate the software development process. GitHub has become a driving force in the digital world and the AI frontier, with numerous tools and libraries hosted by the platform. Its impact is undeniable, and it effectively shapes algorithmic culture and society.

4 Algorithmic Culture and Inclusivity: Who Do These Tools Serve?

GitHub stands out from other similar platforms because of its social features. Although not typically viewed as "social media," GitHub facilitates collaboration and community building. The platform's affordances and usage norms shape community interaction, much like other social media. Acknowledging GitHub's social aspect helps us understand its role in shaping algorithmic culture. Algorithmic culture seeks to describe how culture and technology are fundamentally intertwined. Algorithms shape the information users consume, but human inputs simultaneously influence algorithmic behavior. Human biases, values, culture, and power dynamics are embedded into algorithms. In *Algorithmic Culture Before the Internet,* Striphas explains how technology and culture are fundamentally connected by impact: algorithms organize people, places, and ideas, and our emotions and behavior are shaped by these algorithms (Striphas, 2023).

Like any other social platform, GitHub exhibits a lack of representation and reinforces harmful biases. GitHub is used disproportionately by men, with 91% of users being male. This is not specific to GitHub but rather a broader issue in open-source software (and, more broadly, the tech industry). While anyone can contribute content on GitHub, hierarchies and in-group/out-group dynamics result in systemic exclusion. Some users may not be seen or valued as highly as experienced developers. Women's contributions are often devalued, and they are routinely subjected to discriminatory treatment. A GitHub coding study by researchers at Cal Poly and other universities found that women's code was favored over men's, but only if their gender was unknown ("Github Coding Study Suggests Gender Bias," 2016). Additionally, developers' popularity affects their influence, with users with higher follower counts gaining attention (Asay, 2025). While GitHub's open-source nature suggests freedom of expression for all, the platform is prone to uneven power dynamics and technological biases that dictate whose voices are valued.

5 Detrimental Effects of AI Dependency on Self-Sufficiency and Intellectual Development

AI usage has changed over time and is increasingly being used for coding assistance—to write code and fix errors. In a comparison chart of AI usage purposes in 2024 versus 2025, "Generating code (for pros)" skyrocketed in ranking from 47th to 5th (Zao-Sanders, 2025). Namanyay Goel, an experienced developer, explains how the newest generation of coders heavily depends on AI models to write code, with "Copilot or Claude or GPT running 24/7." They can produce working code efficiently but are entirely unable to understand how it works and its edge cases (Landymore, 2025). AI in coding is an enticing shortcut that eliminates the process of struggling to solve problems but may consequently threaten intellectual capability. Learning foundational algorithms like long division helps you understand more complex ideas intuitively, so getting AI to speed up simpler processes and do the grunt work is problematic. The process of writing code is arguably just as necessary as the end result, as it develops problem-solving, critical thinking, deep understanding, and other invaluable skills.

Though AI is presented as a tool to augment human intelligence, it is increasingly replacing human labor altogether, threatening the future of human employment. AI-powered tools are automating countless roles and replacing human workers. A Forum 2025 report revealed that 40% of employers expect to reduce their workforce in favor of AI (Leopold, 2025). This is quite terrifying, as more and more people struggle to acquire jobs, and the value of human effort and work is diminishing. The threat of AI to the job market is a real concern that should be addressed.

6 AI Development Costs: Energy Overconsumption and Exploitative Human Labor

Billions of dollars, countless resources, and human labor go into AI development, which is marketed enthusiastically and optimistically as a technological revolution. There is little public information on GitHub's energy consumption, but the platform is powered by Codex, a model based on the massive GPT3, which uses an extreme amount of energy (Gautam, 2021). While the company hides the environmental costs of its products, AI development generally involves a massive amount of energy to fuel rapid technological progression. AI training relies on large amounts of data and computationally heavy calculations, requiring significant power and consuming a large amount of fresh water. "Servers are hungry. And thirsty" (Ren, 2023). Corporate incentives prioritize attractive AI advancements, depleting attention and resources that could be directed toward addressing deep social issues. Society is "trapped inside a lopsided imagination of those who monopolize power and resources to benefit the few at the expense of the many" (Benjamin, 2023). Eagerly racing toward an AI-powered future at lightning speed poses a severe systemic threat, as impressive technological progression comes at the direct expense of environmental resources necessary for long-term ecological sustainability.

Human labor is also an essential component of AI development. GitHub's success is largely due to the work of open source contributors, but the platform has faced criticism for unethically monetizing this labor. The company has profited from the work of open-source programmers without pay or credit (Joseph Saveri Law Firm, n.d.). This is not unique to GitHub, as the software development tool Builder.AI "masked manual labor as machine learning," neglecting to document the work of Indian engineers (Jacobs, 2025). This idea is exemplified by the Mechanical Turk, a chess master machine that was actually a human playing chess. This "complex mechanical system … was meant to distract their attention from how the automaton really worked: human labor" (Sadowski, 2018)**.** While AI tools are often prized as autonomous, cutting-edge, and revolutionary advancements, their capacities are entirely dependent upon the underlying, often unseen human labor. When human labor is leveraged without credit or value for corporate benefits, a systemic injustice occurs.

7 Future of AI: Evaluating Long-Term Harms While Developing a Radical Vision Benefiting Humanity

Considering the potential negative impacts of AI on human growth and society, some may form a pessimistic, negative perspective of AI, where it is viewed as society's slayer. GitHub has evident flaws, as it perpetuates harmful biases, exploits the work of programmers, and has environmental costs. Some denounce AI for its tendency to cause intense conflict, competition, and inequality, displacing and harming society. Others view AI as a savior, catering to our needs and desires and knowing us "better than we know ourselves" (Benjamin, 2023). Both narratives put technology in the leading role, taking away our power as humans and only considering technology's impact on humans rather than human impact on technology. The future of AI remains unclear, yet prioritizing human agency in system development and demanding accountability for algorithmic harms are essential steps towards effective, equitable solutions.

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